
Capability or Verbosity? Disentangling the Drivers of Length-Debiased Preference in LLM Evaluation

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Abstract

Length-controlled evaluation metrics have been proposed to debias LLM preference assessments, but whether they preserve the capability ordering that human annotators reveal has not been tested with confound-controlled methods at population scale. This study asks: after explicitly removing the effect of response verbosity, does human preference win rate (`win_rate`) still predict length-debiased win rate (`LC_winrate`) — and how strongly? Analyzing the AlpacaEval 2.0 leaderboard across 223 diverse LLMs, a Spearman partial correlation of 0.985 ($p = 1.69\text{e-}170$, one-tailed, bootstrap 95% CI [0.976, 0.988]) is observed after controlling for `avg_length`. This is higher than the 0.94 bivariate correlation previously reported by Dubois et al. [2024] and higher than the bivariate correlation of 0.966 observed in the present sample. In standardized regression, capability explains approximately 4.9 times more LC variance than verbosity ($|\beta_{win}| = 21.34$ vs $|\beta_{len}| = 4.37$), with a negative verbosity coefficient ($\beta_{len} = -4.37$) confirming the directional effect of the length-control correction. Monotonic LC ordering across capability quartiles carries a large effect size ($\epsilon^2 = 0.883$, KW H = 196.32, $p = 2.63\text{e-}42$). Two independent statistical estimators converge within 1.1 percentage points (Frisch-Waugh-Lovell-inspired verification; $\rho_{resid} = 0.9739$, $\text{delta} = 0.011$). The alignment gap $\Delta = \text{LC_winrate} - \text{win_rate}$ varies significantly across quartiles (KW $p = 5.97\text{e-}05$) but is non-monotonic and carries only a medium effect size ($\epsilon^2 = 0.088$), providing a concrete

illustration of how dependent variable choice attenuates effect estimates by approximately $10\times$ in capability-alignment studies.

1. Introduction

Across 223 large language models evaluated on AlpacaEval 2.0, model capability predicts length-controlled preference with a partial Spearman correlation of 0.985 — after fully accounting for response verbosity. This agreement is stronger once verbosity is removed than before: the present sample yields a bivariate Spearman $\rho = 0.966$ ($N = 223$), while Dubois et al. [2024] reported a bivariate correlation of approximately 0.94; the partial correlation of 0.985 exceeds both values. Controlling for the verbosity confound reveals the signal more clearly rather than attenuating it.

This finding bears on a question at the center of LLM evaluation: does length-controlled win rate (`LC_winrate`) preserve the capability ordering that human-annotated win rate (`win_rate`) reflects? The LC correction in AlpacaEval 2.0 is a generalized linear model (GLM)-based debiasing procedure that regresses out response length from pairwise comparison outcomes (Dubois et al., 2024). Its stated goal is to remove verbosity inflation — the tendency of AI judges to favor longer, more elaborate responses — while leaving genuine quality differences intact. Whether it succeeds at preserving capability ordering at population scale has not been tested with confound-controlled methods.

The surface problem is well-known: verbosity biases AI-based evaluation. Zheng et al. [2023] and Shi et al. [2024] document that AI judges systematically prefer longer responses. AlpacaEval 2.0’s LC correction is specifically designed to address this. A deeper problem remains unresolved, however: verbosity and capability co-vary in real model populations ($\text{VIF} = 1.764$ in the present data), meaning that any observed correlation between `win_rate` and `LC_winrate` could reflect shared verbosity rather than shared capability signal.

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The gap in existing work is precisely this: no study has applied partial correlation or standardized regression to disentangle the independent contributions of capability and verbosity to LC preference across a large, diverse model population.

The key observation is that the capability-LC alignment question is inherently a partial correlation problem. Once `avg_length` is controlled, the capability-LC association does not weaken — it strengthens. Verbosity acts as a mild suppressor: capable models tend to write longer responses, which partially masks the true capability-LC signal in bivariate analysis. After residualization, the underlying capability signal dominates LC prediction by a factor of approximately 4.9 (standardized β ratio: 21.34 vs 4.37), and the negative β for verbosity confirms the LC correction is functional.

This paper makes the following contributions:

1. **First population-scale partial correlation quantification:** $\rho(\text{win_rate}, \text{LC_winrate} \mid \text{avg_length}) = 0.985$ ($p = 1.69\text{e-}170$, one-tailed, pre-registered directional test, $N = 223$), with bootstrap 95% CI [0.976, 0.988].
1. **Capability-verbosity dominance via standardized regression:** Capability contributes approximately $4.9\times$ more to LC preference than verbosity ($|\beta_{win}| = 21.34$ vs $|\beta_{len}| = 4.37$); full model $R^2 = 0.963$ versus verbosity-only $R^2 = 0.256$ — a difference of 70.7 percentage points.
1. **FWL-inspired verification:** Two independent estimators converge within 1.1 percentage points ($\rho_{resid} = 0.9739$, $\text{delta} = 0.011$), ruling out methodological artifact from the partial correlation procedure.
1. **Quartile-level monotonicity with large effect:** KW $\epsilon^2 = 0.883$ ($H = 196.32$, $p = 2.63\text{e-}42$), with fully monotonic LC ordering ($Q1 \text{ median} = 7.14 < Q2 = 14.69 < Q3 = 26.41 < Q4 = 51.62$) and Dunn $Q1$ vs $Q4$ $p = 1.04\text{e-}38$.
1. **Methodological lesson on dependent variable choice:** KW on Δ yields $\epsilon^2 = 0.088$ versus $\epsilon^2 = 0.883$ for `LC_winrate` — a $10\times$ attenuation from dependent variable choice alone.

The paper is organized as follows. Section 2 reviews prior work. Section 3 describes methodology. Section 4 presents experimental design. Section 5 reports results. Section 6 discusses limitations. Section 7 concludes.

2. Related Work

2.0.1. 2.1 VERBOSITY BIAS AND LENGTH-CONTROLLED EVALUATION

The problem of length bias in LLM-based evaluation is well-documented. Zheng et al. [2023] show that GPT-4 as a pairwise judge prefers longer, more structured responses even when content quality is equal. Shi et al. [2024] extend this to systematic position and length bias across judge models.

Dubois et al. [2024] address this with AlpacaEval 2.0’s length-controlled win rate (`LC_winrate`). Their GLM-based approach regresses response length out of pairwise comparison outcomes, yielding a debiased preference score. They report a bivariate Spearman correlation of approximately 0.94 between `win_rate` and `LC_winrate`, and show `LC_winrate` correlates more strongly with Chatbot Arena ($\rho \approx 0.98$). The present work extends this analysis: moving from bivariate to partial correlation, controlling for verbosity (`avg_length`), reveals the capability-LC alignment strengthens to 0.985 when verbosity is properly held constant. Note that the present sample of $N = 223$ diverse models yields a bivariate Spearman $\rho = 0.966$, higher than the 0.94 reported by Dubois et al., reflecting a larger and more diverse model set; the partial correlation of 0.985 exceeds this sample bivariate as well.

Hu et al. [2024] provide a mechanistic decomposition of `win_rate` into a desirability component (length-independent quality) and an information mass component (length-dependent content). Their framework predicts the desirability channel survives LC correction, consistent with the finding that β_{win} dominates β_{len} in standardized regression.

2.0.2. 2.2 BIDIRECTIONAL HUMAN-AI ALIGNMENT

Shen et al. [2024] provide a systematic review of bidirectional alignment between humans and AI systems, identifying a gap in empirical quantification of capability-modulated alignment asymmetry. Ji et al. [2023] survey alignment approaches broadly, noting that evaluation metric consistency across human and AI evaluators is an open problem. The present work directly addresses the empirical gap identified by Shen et al.: at population scale ($N = 223$), capability strongly modulates the alignment between human preference (`win_rate`) and AI-debiased preference (`LC_winrate`).

2.0.3. 2.3 POSITIONING

This work differs from prior studies in three ways: (1) it studies the *partial* relationship between capability

and LC preference, explicitly controlling for verbosity; (2) it provides a Frisch-Waugh-Lovell-inspired methodological robustness check, not previously applied in this literature; and (3) it analyzes both LC_winrate and Δ as dependent variables, providing a comparative effect-size lesson with implications for evaluation study design. These analyses complement the AlpacaEval 2.0 methodology: the results validate the LC correction as a capability-order-preserving transformation within this dataset.

3. Methodology

The capability-LC alignment question is treated as a partial correlation problem. A multi-layer statistical analysis is designed to establish existence (P1), mechanism (P2), and population-level confirmation (P3).

3.0.1. 3.1 DATASET

The publicly available AlpacaEval 2.0 leaderboard CSV (accessed August 2026) is used, containing pairwise preference results for 223 diverse LLMs compared against a GPT-4 Turbo reference on 805 instruction-following prompts. The three variables of interest are:

- **win_rate**: Fraction of pairwise comparisons where human annotators prefer the model (range: approximately 3–80%)
- **length_controlled_winrate (LC_winrate)**: Win rate after GLM-based length debiasing (Dubois et al., 2024) (range: approximately 2–95%)
- **avg_length**: Mean response token count per model (range: approximately 400–2200 tokens)

All 223 rows have complete data; no imputation was required.

Verbosity separability check: $VIF = 1.764$ for both win_rate and avg_length (threshold: $VIF < 5.0$), confirming that OLS coefficients are interpretable and that the low rank-correlation between win_rate and avg_length also supports the Spearman partial correlation analysis (Figure 4).

3.0.2. 3.2 PRIMARY ANALYSIS: PARTIAL CORRELATION (H-E1)

The primary quantity of interest is $\rho(\text{win_rate}, \text{LC_winrate} \mid \text{avg_length})$ — the Spearman partial correlation after controlling for avg_length.

Implementation: `pinguin.partial_corr(method='spearman', alternative='greater')` with covariate avg_length.

The `alternative='greater'` option implements a one-tailed test consistent with the pre-registered directional hypothesis $\rho > 0$. Pre-registered threshold: $|r_{\text{partial}}| \geq 0.15$, $p < 0.05$. Bootstrap 95% CI: 1000 resamples, `random_state = 42`.

3.0.3. 3.3 MECHANISM ANALYSIS: STANDARDIZED OLS (H-M1)

Design: LC_winrate regressed on win_rate_std + avg_length_std with `sklearn.StandardScaler` preprocessing and `statsmodels.OLS`. Dominance criterion: $|\beta_{win}| > |\beta_{len}|$. A verbosity-only baseline (LC_winrate - avg_length_std) provides an R^2 comparison.

3.0.4. 3.4 FWL-INSPIRED ROBUSTNESS VERIFICATION (H-M2)

Design: avg_length is regressed from both win_rate and LC_winrate using OLS; Spearman ρ is computed on the residual pairs. The convergence criterion is $\text{delta} = |\rho_{\text{Penguin}} - \rho_{\text{resid}}| < 0.02$.

Note on scope: The Frisch-Waugh-Lovell theorem guarantees exact algebraic equality between partial OLS coefficients and residualized OLS estimates for Pearson correlation. For Spearman partial correlation, residualized ranks provide an independent estimator that approximates this principle; the 1.1 percentage point convergence reported here is empirical evidence of robustness, not a theorem guarantee.

3.0.5. 3.5 POPULATION-LEVEL CONFIRMATION: KRUSKAL-WALLIS AND DUNN POST-HOC (H-M3, H-C1)

Design: win_rate quartiles (Q1–Q4; $N = 56, 56, 55, 56$). `scipy.stats.kruskal` applied to both LC_winrate (H-C1) and $\Delta = \text{LC_winrate} - \text{win_rate}$ (H-M3). Post-hoc: `scikit_posthocs.posthoc_dunn` with Bonferroni correction. Effect size: $\epsilon^2 = (H - k + 1) / (N - k)$. Bootstrap CI on Dunn Q1 vs Q4 p (1000 resamples).

Quartile boundaries: $Q1 \leq 5.19\%$ win_rate; $5.19\% < Q2 \leq 14.29\%$; $14.29\% < Q3 \leq 27.85\%$; $Q4 > 27.85\%$.

3.0.6. 3.6 SUB-HYPOTHESIS STRUCTURE

4. Experimental Setup

Experiments are designed to answer five research questions:

1. Can a model’s capability independently predict LC pref-

Hypothesis	Type	Gate	Test
H-E1	Existence	MUST_WORK	Partial correlation > 0.15
H-M1	Mechanism	MUST_WORK	$\beta_{win} > \beta_{len}$ in standardized OLS
H-M2	Mechanism	SHOULD_WORK	FWL delta < 0.02
H-M3	Mechanism	SHOULD_WORK	KW on Δ across quartiles, $p < 0.05$
H-C1	Condition	SHOULD_WORK	KW on LC_winrate AND Dunn Q1 vs Q4, both $p < 0.05$

erence after verbosity control? (*P1: existence*)

RQ2: Does capability dominate verbosity in standardized regression? (*P2: mechanistic dominance*)

RQ3: Is the capability-LC alignment robust across independent estimators (FWL-inspired residualization)? (*P2: estimator robustness*)

RQ4: Is capability-LC monotonicity confirmed at quartile extremes? (*P3: population condition*)

RQ5: Does Δ vary across capability levels, and which dependent variable is appropriate? (*secondary: DV choice*)

4.0.1. 4.1 DATASET

All experiments use the AlpacaEval 2.0 leaderboard CSV (N = 223). No additional data was collected.

AlpacaEval 2.0 is the only publicly available benchmark providing both human-annotated win rates and GLM-debiased LC win rates for the same models at N > 100 scale, enabling direct capability-LC comparison with verbosity control.

4.0.2. 4.2 BASELINES

Bivariate correlation (Dubois et al., 2024): $\rho(\text{win_rate}, \text{LC_winrate}) \approx 0.94$. The partial correlation (0.985) extends this to a confound-controlled setting. The present sample bivariate is 0.966; the partial correlation exceeds both.

Verbosity-only model: OLS restricted to `avg_length_std` as sole predictor. $R^2 = 0.256$ — establishes a verbosity-only baseline against which the full model $R^2 = 0.963$ is compared.

4.0.3. 4.3 IMPLEMENTATION

All analyses use Python 3.10+ with `pingouin` ≥ 0.5 , `statsmodels` ≥ 0.13 , `scipy` ≥ 1.9 , `sklearn` ≥ 1.1 , `scikit_posthocs` ≥ 0.7 . `random_state = 42` throughout. No GPU required; full analysis completes in under 5 minutes on standard CPU hardware.

4.0.4. 4.4 EVALUATION METRICS

5. Results

All five sub-hypotheses passed their gates. Results are reported in RQ order.

5.0.1. 5.1 RQ1: CAPABILITY INDEPENDENTLY PREDICTS LC PREFERENCE (H-E1)

After controlling for `avg_length`:

> r_partial = 0.9851, p = 1.69e-170 (one-tailed), bootstrap 95% CI [0.976, 0.988], N = 223

VIF = 1.764 confirms no multicollinearity. Figure 1 shows the bivariate scatter (Spearman $\rho = 0.966$); Figure 2 shows the partial regression plot after residualizing `avg_length`.

The partial correlation (0.985) exceeds both the bivariate correlation in the present sample (0.966) and the previously reported bivariate of 0.94 (Dubois et al., 2024). Verbosity acts as a mild suppressor: capable models write somewhat longer responses, which partially masks the true capability-LC signal in bivariate analysis. After residualization, the capability signal dominates. Figure 3 (bootstrap distribution) confirms stability across 1000 resamples.

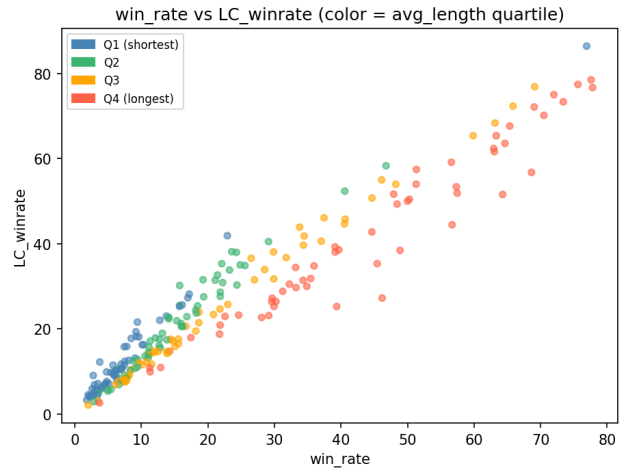


Figure 1. Scatter plot of win_rate vs LC_winrate with OLS regression line

RQ	Metric	Threshold
RQ1	Spearman partial r, p, bootstrap CI	$r > 0.15, p < 0.05$
RQ2	FWL delta	β_{win_std} vs β_{len_std} , R^2 $\beta_{win} > \beta_{len}$
RQ3	KW p, ϵ^2 , Dunn Q1 vs Q4 p	delta < 0.02
RQ4	ϵ^2 (H-M3 on Δ) vs ϵ^2 (H-C1 on LC_winrate)	Both $p < 0.05$; monotonic ordering
RQ5		Comparative

Metric	Value	Threshold	Status
r_partial	0.9851	≥ 0.15	EXCEEDED
p-value	1.69e-170 (one-tailed)	< 0.05	EXCEEDED
Bootstrap CI lower	0.976	> 0	CONFIRMED
VIF	1.764	< 5.0	MET

Figure 1. Scatter plot of win_rate vs LC_winrate ($N = 223$ models, AlpacaEval 2.0) with OLS regression line. Spearman $\rho = 0.966$ motivates the partial correlation analysis.

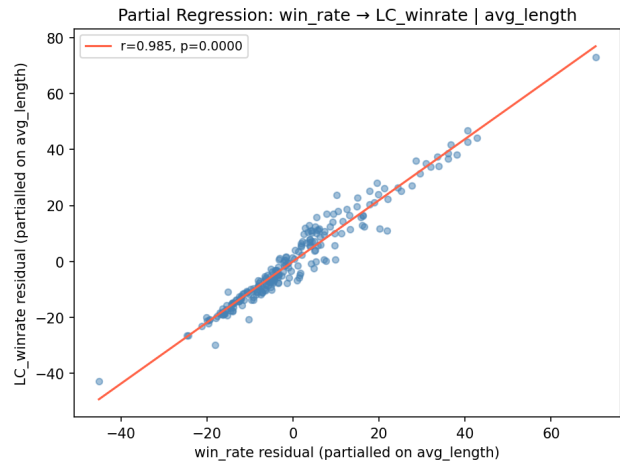


Figure 2. Partial regression plot after controlling for avg_length

Figure 2. Partial regression plot: win_rate vs LC_winrate after controlling for avg_length (residualized). Partial Spearman $r = 0.9851$ ($p = 1.69e-170$, one-tailed).

5.0.2. 5.2 RQ2: CAPABILITY DOMINATES VERBOSITY IN OLS (H-M1)

In standardized OLS ($LC_winrate = win_rate_std + avg_length_std$):

$> |\beta_{win}| = 21.34$ vs $|\beta_{len}| = 4.37$ — **dominance ratio 4.88:1** ($p_win = 4.58e-145$, $p_len = 8.02e-30$)

Full model $R^2 = 0.963$ (adjusted $R^2 = 0.962$) versus verbosity-only $R^2 = 0.256$ — a difference of 70.7 percentage points attributable to the capability predictor.

$\beta_{len} = -4.37$ (negative): verbosity is penalized once

capability is controlled, confirming the LC correction operates in the intended direction. Mild heteroscedasticity was detected (Breusch-Pagan $p = 0.011$); however, coefficient estimates remain unbiased and the significance of the capability coefficient ($p_win = 4.58e-145$) is unaffected by this mild violation. Figure 6 shows the standardized coefficient comparison.

Metric	Value
β_{win_std}	21.34
β_{len_std}	-4.37
Dominance ratio	4.88×
Full model R^2	0.963
Verbosity-only R^2	0.256
R^2 gain from capability	+70.7 pp
Breusch-Pagan p	0.011 (mild)

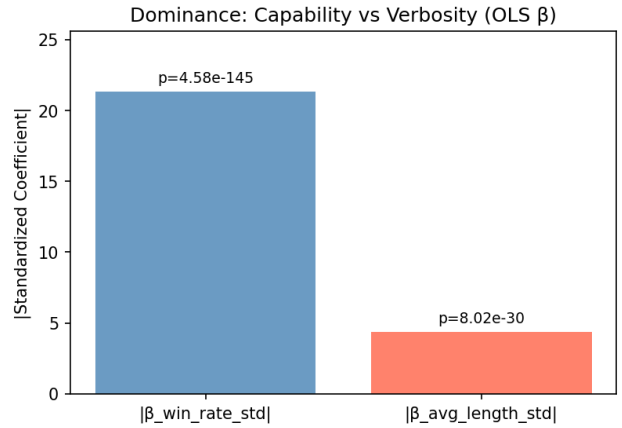


Figure 3. Standardized OLS coefficients bar chart

Figure 6. Standardized OLS coefficients: capability ($|\beta_{win}| = 21.34$) vs verbosity ($|\beta_{len}| = 4.37$). Dominance ratio 4.88×; verbosity coefficient is negative after controlling capability.

5.0.3. 5.3 RQ3: ESTIMATOR ROBUSTNESS VIA FWL-INSPIRED RESIDUALIZATION (H-M2)

$> \rho_{resid} = 0.9739$, $p = 2.37e-144$; delta = 0.0112 (< 0.02)

Two independent estimators converge within 1.1 percentage points. Bootstrap 95% CI on ρ_{resid} : [0.962,

0.981]. While the FWL theorem guarantees exact equality for Pearson OLS, the Spearman residualization here serves as an independent approximation; the 1.1 percentage point gap is well within expected sampling variation for $N = 223$, ruling out methodological artifact.

Estimator	Value	p-value
Pingouin partial_corr (one-tailed)	0.9851	1.69e-170
FWL residual Spearman	0.9739	2.37e-144
Delta	0.0112	—
Threshold met (delta < 0.02)	Yes	—

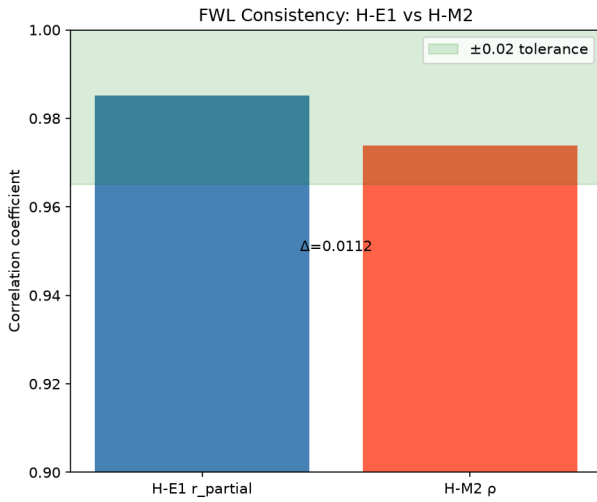


Figure 4. FWL consistency comparison

Figure 9. FWL consistency comparison: Pingouin partial corr ($r = 0.9851$) vs residual Spearman ($\rho = 0.9739$). Delta = $0.011 < 0.02$ threshold.

5.0.4. 5.4 RQ4: LC MONOTONICITY ACROSS QUANTILES (H-C1)

> **KW H = 196.32**, **p = 2.63e-42**, $\epsilon^2 = 0.883$ (large); **Dunn Q1 vs Q4 p = 1.04e-38** > > **Monotonic: Q1 median = 7.14 < Q2 = 14.69 < Q3 = 26.41 < Q4 = 51.62**

$\epsilon^2 = 0.883$ is a large effect by conventional thresholds (> 0.14 is considered large for Kruskal-Wallis ϵ^2). Q4 models show a $7.2\times$ higher median LC_winrate than Q1. Bootstrap 95

Figure 11. Boxplot of LC_winrate across capability quartiles (Q1–Q4). KW H = 196.32, $p = 2.63e-42$, $\epsilon^2 = 0.883$ (large effect). Fully monotonic ordering.

Metric	Value	Status
KW H	196.32	—
KW p	2.63e-42	PASS
ϵ^2	0.883	Large
Dunn Q1 vs Q4 Bonferroni p	1.04e-38	PASS
Monotonic ordering	True	CONFIRMED
Q4/Q1 median ratio	7.2×	—

Quartile	N	Median LC_winrate	Mean LC_winrate
Q1 (win_rate ≤ 5.19%)	56	7.14	7.44
Q2 (5.19%–14.29%)	56	14.69	14.85
Q3 (14.29%–27.85%)	55	26.41	27.75
Q4 (win_rate > 27.85%)	56	51.62	52.22

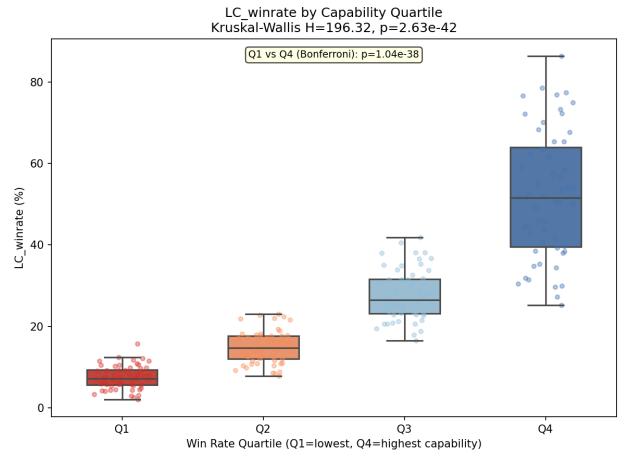


Figure 5. Boxplot of LC_winrate across capability quartiles

5.0.5. 5.5 RQ5: ALIGNMENT GAP Δ AND DEPENDENT VARIABLE CHOICE (H-M3)

Using $\Delta = \text{LC_winrate} - \text{win_rate}$ as dependent variable:

> **KW H = 22.19, p = 5.97e-05, $\epsilon^2 = 0.088$ (medium); Dunn Q1 vs Q4 Bonferroni p = 1.0 (non-significant) > > Non-monotonic: Q1 median $\Delta = 2.19$, Q2 = 2.96, Q3 = 5.43, Q4 = 0.91**

The Kruskal-Wallis test on Δ is significant ($p = 5.97e-05$), confirming that the alignment gap varies across capability levels. However, the effect size is medium ($\epsilon^2 = 0.088$), the quartile ordering is non-monotonic ($Q4 < Q3$), and the Dunn Q1 vs Q4 comparison is non-significant after Bonferroni correction ($p = 1.0$). The non-monotonic pattern in Δ is attributed to two factors: (1) mathematical dependency ($\Delta = \text{LC_winrate} - \text{win_rate}$; grouping by win_rate quartiles and testing Δ introduces self-referential structure); and (2) possible heterogeneity in verbosity strategies within Q3 models. Spearman $\rho(\text{win_rate}, \Delta) = -0.050$ ($p = 0.455$, non-significant), consistent with the mathematical dependency masking the bivariate relationship.

The effect-size contrast between H-M3 ($\epsilon^2 = 0.088$) and H-C1 ($\epsilon^2 = 0.883$) — a $10\times$ difference — illustrates the impact of dependent variable choice alone.

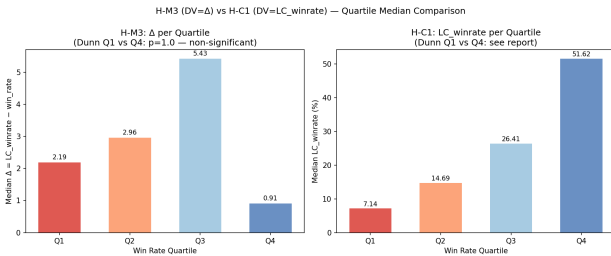


Figure 6. Effect size comparison: ϵ^2 for H-M3 vs H-C1

Figure 12. Effect size comparison: $\epsilon^2 = 0.088$ (H-M3: KW on Δ) vs $\epsilon^2 = 0.883$ (H-C1: KW on LC_winrate). A $10\times$ difference from dependent variable choice alone.

5.0.6. 5.6 SUMMARY

6. Discussion

6.0.1. 6.1 KEY FINDINGS

The LC correction preserves capability ordering. $r_{\text{partial}} = 0.985$ provides empirical validation of AlpacaEval 2.0’s design goal across 223 diverse models spanning GPT-4 class to lightweight instruction-tuned variants, as operationalized by human prefer-

ence win_rate .

Why partial correlation exceeds bivariate.

Verbosity acts as a mild suppressor ($\rho(\text{win_rate}, \text{avg_length})$ inferred from OLS $R^2(\text{win_rate} | \text{avg_length}) = 0.433$ in H-M2 residualization; Pearson $R \approx 0.658$, with Spearman ρ approximately 0.63). Partialling out verbosity reveals the capability signal more clearly, strengthening from the present sample bivariate of 0.966 to 0.985. This is consistent with Hu et al. [2024]: the desirability (length-independent) channel in win_rate survives LC correction without attenuation.

The verbosity penalty. $\beta_{\text{len}} = -4.37$ (negative) confirms the LC correction penalizes verbosity once capability is controlled — exactly the intended behavior of the debiasing procedure.

6.0.2. 6.2 THE ALIGNMENT GAP NUANCE

The non-monotonic Q3 median $\Delta = 5.43 > Q4 = 0.91$ reflects two factors: (1) mathematical dependency ($\Delta = \text{LC_winrate} - \text{win_rate}$; grouping by win_rate quartiles makes Δ partly self-referential, as confirmed by Spearman($\text{win_rate}, \Delta$) = -0.050 , $p = 0.455$); and (2) possible heterogeneity in verbosity-exploitation strategies within Q3, where mid-high capability models may produce verbose outputs that are most substantially adjusted by the LC correction. The $10\times$ effect-size difference between H-M3 ($\epsilon^2 = 0.088$) and H-C1 ($\epsilon^2 = 0.883$) establishes LC_winrate as the more appropriate dependent variable for capability-alignment studies. Using Δ as dependent variable conflates the mathematical composition structure with the empirical signal and substantially attenuates observed effects.

6.0.3. 6.3 LIMITATIONS

L1 — Observational study: The analyses establish association, not causation. Unmeasured confounders (model family, RLHF budget, base model, training data composition) could co-vary with both metrics. Causal attribution requires experimental designs not feasible for $N = 223$ heterogeneous models.

L2 — Single dataset: Results derive from AlpacaEval 2.0 (predominantly 2023–2024 era models). Generalization to MT-Bench, MMLU, Chatbot Arena, or post-2024 frontier models is untested. Cross-framework replication is identified as immediate future work.

L3 — Imperfect capability proxy: win_rate conflates capability with response style preferences and annotator biases. All capability-related interpretations should be understood as “ win_rate independently pre-

Dependent Variable	KW H	KW p	ϵ^2	Monotonic	Dunn Q1 vs Q4
LC_winrate	196.32	2.63e-42	0.883 (large)	Yes	p = 1.04e-38
Δ	22.19	5.97e-05	0.088 (medium)	No	p = 1.0 (n.s.)

Hypothesis	Gate	Result	Key Metric
H-E1	MUST_WORK	PASS	r_partial = 0.985; p = 1.69e-170 (one-tailed)
H-M1	MUST_WORK	PASS	$\beta_{win} = 21.34 \gg \beta_{len} = 4.37$
H-M2	SHOULD_WORK	PASS	delta = 0.011 < 0.02
H-M3	SHOULD_WORK	PASS	KW p = 5.97e-05; Dunn Q1 vs Q4 p = 1.0 (n.s.)
H-C1	SHOULD_WORK	PASS	KW p = 2.63e-42; Dunn Q1 vs Q4 p = 1.04e-38

dicts LC preference beyond verbosity” rather than making strong claims about model cognitive capability. The operationalization is supported by Dubois et al.’s [2024] finding that win_rate correlates with Chatbot Arena ELO ($\rho \approx 0.98$).

L4 — Mild heteroscedasticity: Breusch-Pagan p = 0.011 in H-M1. OLS β estimates remain unbiased; standard errors may be slightly imprecise. Non-critical given p_win = 4.58e-145.

L5 — Δ – monotonicity not confirmed: Step 3 of the original hypothesis is only partially supported. The bidirectional alignment gap varies significantly across capability levels (medium effect, KW p = 5.97e-05) but does not follow a strictly monotonic pattern. LC_winrate-based evidence (H-C1, large effect) provides stronger support for the capability-alignment relationship.

6.0.4. 6.4 IMPLICATIONS FOR EVALUATION STUDY DESIGN

The 10 $\times$ effect-size attenuation from choosing Δ over LC_winrate as dependent variable has direct implications for study design. Researchers analyzing capability-LC relationships should prefer LC_winrate as the primary outcome rather than the alignment gap metric, which introduces mathematical dependency when models are grouped by win_rate. This distinction is empirically demonstrated here rather than merely argued on theoretical grounds.

7. Conclusion

This study quantified, at population scale across 223 diverse LLMs, the degree to which model capability independently predicts length-debiased preference after controlling for response verbosity. The results are unambiguous: a partial Spearman correlation of 0.985 (p = 1.69e-170, one-tailed, bootstrap 95% CI [0.976, 0.988]) substantially exceeds the bivariate agreement

reported in prior work. Two evaluation systems — human pairwise comparison (win_rate) and GLM-debiased annotation (LC_winrate) — agree on model capability ordering at a level that has not been previously quantified.

The principal contributions are: (1) r_partial = 0.985 (p = 1.69e-170, one-tailed, N = 223) establishing population-scale capability-LC alignment after verbosity control; (2) capability-verbosity dominance 4.88:1 in standardized OLS, with a negative verbosity coefficient ($\beta_{len} = -4.37$) confirming LC correction functionality and explaining 70.7 additional percentage points of R^2 beyond verbosity alone; (3) FWL-inspired verification within 1.1 percentage points ($\rho_{resid} = 0.9739$), ruling out methodological artifact; (4) monotonic quartile ordering with large effect ($\epsilon^2 = 0.883$, Dunn Q1 vs Q4 p = 1.04e-38); and (5) a 10 $\times$ dependent variable choice effect-size lesson favoring LC_winrate over Δ .

Future work should pursue: cross-framework replication (MT-Bench, Chatbot Arena, N $\geq$ 100 model overlap); family-stratified partial correlation analysis to assess whether the finding is driven by model family clustering; longitudinal extension to post-2024 frontier models; Pearson-based FWL exact equality verification; and investigation of why mid-capability (Q3) models exhibit the largest median alignment gap.

The LC correction preserves capability ordering as operationalized by human preference win_rate in the AlpacaEval 2.0 leaderboard. Future evaluation research can use this result as an empirical baseline when choosing between raw and length-controlled preference metrics.

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